

1. The graph of g consists of two straight lines and a semi-circle. Use it to evaluate each integral.

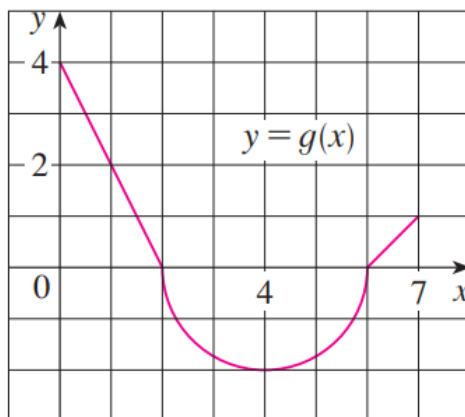
a) $\int_0^2 g(x) dx$

b) $\int_2^6 g(x) dx$

c) $\int_0^7 g(x) dx$

d) $\int_5^5 g(x) dx$

e) $\int_4^1 g(x) dx$



2. Suppose that $f(x)$ & $g(x)$ are continuous functions at all real numbers and that

$$\int_0^1 f(x) dx = 2 \quad \int_0^2 f(x) dx = -1 \quad \int_0^1 g(x) dx = 1 \quad \int_0^2 g(x) dx = 0$$

Use properties of integrals to compute each of the following.

a) $\int_0^1 [3f(x) + 2g(x)] dx$

b) $\int_1^2 g(x) dx$

c) $\int_1^0 f(x) dx$

d) $\int_1^2 [3 + 2f(x) - 5g(x)] dx$

3. Which of the following is true? How can you fully justify your answer without the use of a calculator?

A. $\int_0^1 \sqrt{1+x^2} dx \geq \int_0^1 \sqrt{1+x} dx$ OR B. $\int_0^1 \sqrt{1+x^2} dx \leq \int_0^1 \sqrt{1+x} dx$

4. Computing the integral $\int_0^4 \frac{\pi}{16\pi + 8\arctan(x^2 e^x)} dx$ is very difficult to do. Using properties of integrals to determine lower and upper bounds on the value of this integral.

5. Suppose that $f(x)$ is a continuous function and the average value of $f(x)$ on $[a, 2a]$ is -10 (where a is some non-zero real number).

a) Show that the area between $f(x)$ and the x -axis on this same interval is $-10a$.

b) If we know that $\int_a^0 f(x) dx = -5a - a^2$ determine a non-zero value of a such that $\int_0^{2a} f(x) dx = 0$.

c) Show that the function $f(x)$ must have a root on the interval $[0, 10]$.